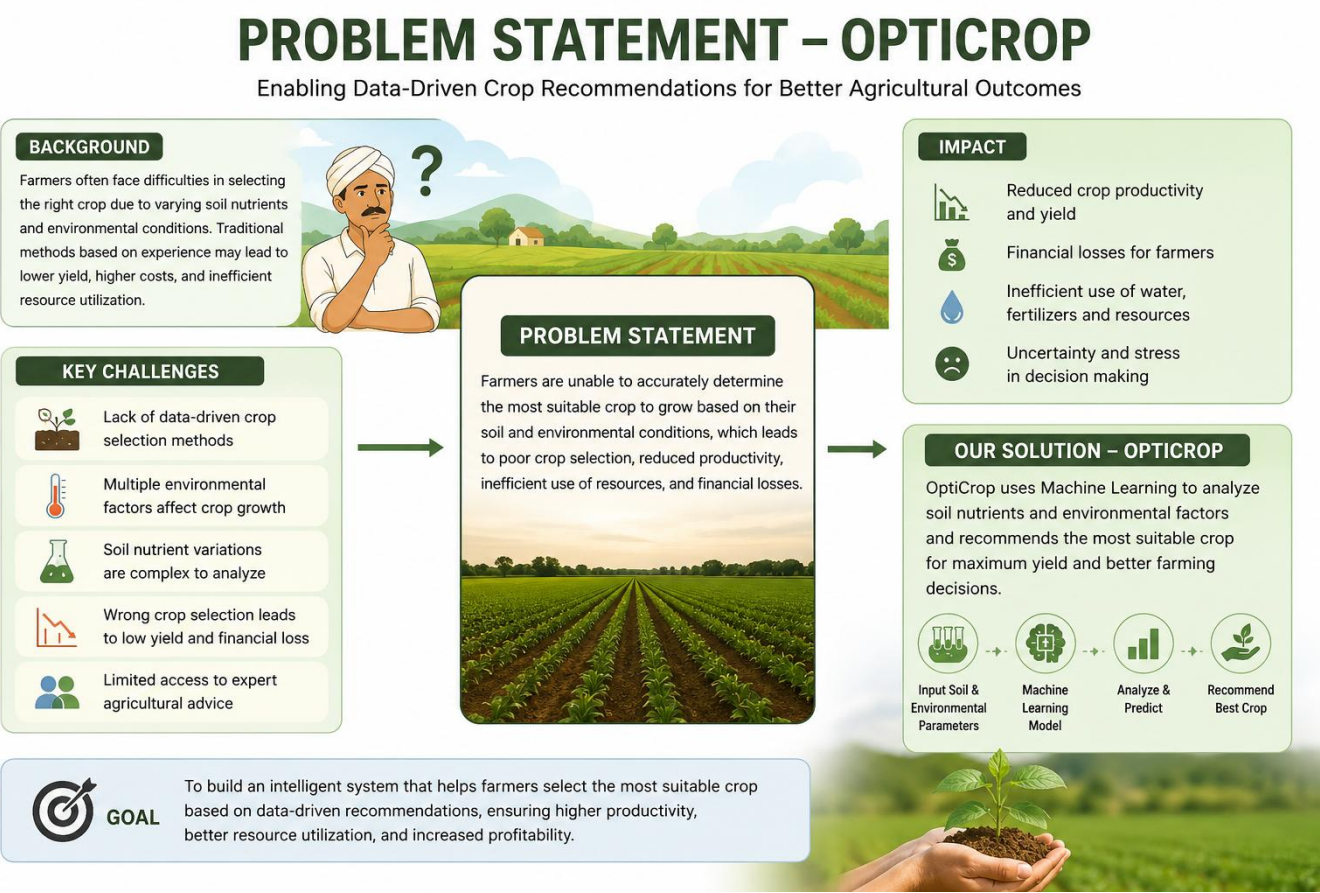


## Project Initialization and Planning Phase

|               |            |
|---------------|------------|
| Date          | 29-06-2026 |
| Team ID       | XXXXXX     |
| Project Name  | Opticrop   |
| Maximum Marks | 3 Marks    |

### Define Problem Statements (Customer Problem Statement Template):

Many farmers struggle to identify the most suitable crop for cultivation because crop selection depends on multiple factors such as soil nutrient levels, temperature, humidity, pH, and rainfall. Due to limited access to data-driven decision-making tools, farmers often rely on traditional practices or assumptions, which can lead to reduced productivity, inefficient resource utilization, and financial losses. There is a need for an intelligent system that can analyze soil and environmental conditions and recommend the most appropriate crop for cultivation.



| <b>Problem Statement (PS)</b> | <b>I am (Customer)</b>                             | <b>I'm trying to</b>                                             | <b>But</b>                                                                             | <b>Because</b>                                                                                              | <b>Which makes me feel</b>                                        |
|-------------------------------|----------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| PS-1                          | A farmer planning for the next cultivation season. | Select the most suitable crop for my land.                       | I am unsure which crop will grow best under current soil and environmental conditions. | Crop selection depends on multiple factors such as soil nutrients, temperature, humidity, pH, and rainfall. | Optimistic about loan approval.                                   |
| PS-2                          | An agricultural consultant.                        | Provide accurate crop recommendations to farmers.                | It is difficult to manually analyze multiple soil and environmental parameters.        | Traditional analysis methods are time-consuming and may not always be accurate.                             | Concerned about providing reliable recommendations.               |
| PS-3                          | A farming student or researcher.                   | Study crop suitability under different environmental conditions. | I lack a system that can quickly analyze agricultural data and provide predictions.    | Agricultural data contains complex relationships that are difficult to interpret manually.                  | Frustrated by the time required for analysis and experimentation. |